身份、组织与审计

门店
store

PK	**store_id**	BIGINT
UQ	store_number	VARCHAR(30)
	store_name	VARCHAR(120)
	address ?	VARCHAR(255)
	business_status	VARCHAR(20)
	opening_time	TIME
	closing_time	TIME

员工
employee

PK	**employee_id**	BIGINT
UQ	employee_number	VARCHAR(30)
	full_name	VARCHAR(100)
UQ	email	VARCHAR(255)
	phone ?	VARCHAR(40)
	business_status	VARCHAR(20)
	address ?	VARCHAR(255)
	note ?	VARCHAR(500)

账号
account

PK	**account_id**	BIGINT
UQ	login_name	VARCHAR(100)
	password_hash	VARCHAR(255)
	enabled	BOOLEAN
FK UQ	customer_id ?	BIGINT > customer
FK UQ	employee_id ?	BIGINT > employee
FK UQ	contractor_id ?	BIGINT > repair_contractor

角色授权
role_grant

PK	**role_grant_id**	BIGINT
FK	account_id	BIGINT > account
	role_code	VARCHAR(30)
FK	store_id ?	BIGINT > store
	valid_from	DATETIME
	valid_to ?	DATETIME

关键操作审计
audit_event

PK	**audit_event_id**	BIGINT
FK	actor_account_id	BIGINT > account
	object_type	VARCHAR(40)
	object_id	BIGINT
	action_code	VARCHAR(40)
	occurred_at	DATETIME
	result_code	VARCHAR(20)
	detail_summary ?	VARCHAR(1000)

顾客与租赁需求

顾客
customer

PK	**customer_id**	BIGINT
UQ	customer_number	VARCHAR(30)
	full_name	VARCHAR(100)
UQ	email	VARCHAR(255)
	phone	VARCHAR(40)
	address ?	VARCHAR(255)
	birth_year ?	SMALLINT
	height_cm ?	DECIMAL(5,2)
	weight_kg ?	DECIMAL(5,2)
	boot_size ?	DECIMAL(4,1)
	height_estimated	BOOLEAN
	weight_estimated	BOOLEAN
	boot_size_estimated	BOOLEAN

租赁需求
rental_request

PK	**request_id**	BIGINT
UQ	request_number	VARCHAR(30)
FK	customer_id	BIGINT > customer
FK	service_store_id	BIGINT > store
FK	accepted_by_employee_id ?	BIGINT > employee
	rental_start_date	DATE
	rental_days	INT
	skill_level	VARCHAR(20)
	submitted_at	DATETIME
	request_status	VARCHAR(20)

需求项目
rental_request_item

PK	**request_item_id**	BIGINT
FK	request_id	BIGINT > rental_request
	equipment_category	VARCHAR(40)
	quantity	INT
	fitting_note ?	VARCHAR(255)

需求补充历史
rental_request_supplement

PK	**supplement_id**	BIGINT
FK	request_id	BIGINT > rental_request
	cycle_number	INT
FK	requested_by_employee_id	BIGINT > employee
	reason	VARCHAR(500)
	requested_at	DATETIME
	customer_response ?	VARCHAR(500)
	responded_at ?	DATETIME

UQ (request_id, cycle_number)

租赁合同与调整

租赁合同
rental_contract

PK	**contract_id**	BIGINT
UQ	contract_number	VARCHAR(30)
FK UQ	source_request_id ?	BIGINT > rental_request
FK	customer_id	BIGINT > customer
FK	created_by_employee_id	BIGINT > employee
FK	service_store_id	BIGINT > store
	contract_date	DATE
	rental_start_date	DATE
	rental_days	INT
	amount_due	DECIMAL(12,2)
	contract_status	VARCHAR(20)

合同项目
rental_item

PK	**rental_item_id**	BIGINT
FK	contract_id	BIGINT > rental_contract
FK	equipment_id	BIGINT > equipment
	daily_rate_snapshot	DECIMAL(12,2)
	checked_out_at ?	DATETIME
	returned_at ?	DATETIME
FK	return_store_id ?	BIGINT > store
	item_status	VARCHAR(20)

租赁调整
rental_change

PK	**rental_change_id**	BIGINT
UQ	change_number	VARCHAR(30)
FK	contract_id	BIGINT > rental_contract
	change_type	VARCHAR(20)
	change_reason ?	VARCHAR(255)
	effective_date ?	DATE
	amount_delta ?	DECIMAL(12,2)
FK	handled_by_employee_id ?	BIGINT > employee
	changed_at ?	DATETIME
	change_status	VARCHAR(20)
FK	requested_by_customer_id ?	BIGINT > customer
	requested_at ?	DATETIME
FK	requested_store_id ?	BIGINT > store
	request_note ?	VARCHAR(500)

调整项目关系
rental_change_item

PK	**rental_change_item_id**	BIGINT
FK	rental_change_id	BIGINT > rental_change
FK	rental_item_id	BIGINT > rental_item
	effect_code	VARCHAR(20)

UQ (rental_change_id, rental_item_id, effect_code)

装备与检查

滑雪装备
equipment

PK	**equipment_id**	BIGINT
UQ	inventory_number	VARCHAR(40)
	equipment_name	VARCHAR(120)
	description ?	VARCHAR(500)
	equipment_category	VARCHAR(40)
	skill_level	VARCHAR(20)
FK	home_store_id	BIGINT > store
FK	current_store_id	BIGINT > store
	equipment_status	VARCHAR(20)
	purchase_date ?	DATE
	purchase_price ?	DECIMAL(12,2)
	current_daily_rate	DECIMAL(12,2)
	retail_price ?	DECIMAL(12,2)

装备检查
equipment_inspection

PK	**inspection_id**	BIGINT
FK	equipment_id	BIGINT > equipment
FK	inspected_by_employee_id	BIGINT > employee
FK	rental_item_id ?	BIGINT > rental_item
FK	repair_task_id ?	BIGINT > repair_task
	inspection_type	VARCHAR(20)
	inspected_at	DATETIME
	inspection_result	VARCHAR(20)
	fault_description ?	VARCHAR(500)

租赁费用与交易结果

第三方交易结果
payment_transaction

PK	**payment_transaction_id**	BIGINT
FK	rental_charge_id	BIGINT > rental_charge
	transaction_type	VARCHAR(20)
UQ	external_transaction_id	VARCHAR(100)
	amount	DECIMAL(12,2)
	transaction_status	VARCHAR(20)
	processed_at ?	DATETIME
FK	recorded_by_employee_id	BIGINT > employee
	recorded_at	DATETIME

租赁费用
rental_charge

PK	**rental_charge_id**	BIGINT
FK	contract_id	BIGINT > rental_contract
FK	rental_change_id ?	BIGINT > rental_change
	charge_type	VARCHAR(20)
	amount	DECIMAL(12,2)
	charge_status	VARCHAR(20)
	created_at	DATETIME

维修任务与承包商

维修任务
repair_task

PK	**repair_task_id**	BIGINT
UQ	repair_task_number	VARCHAR(30)
FK	equipment_id	BIGINT > equipment
FK	requested_by_employee_id	BIGINT > employee
FK	service_store_id	BIGINT > store
FK	current_contractor_id ?	BIGINT > repair_contractor
	request_date	DATE
	fault_description	VARCHAR(500)
	handling_requirement ?	VARCHAR(500)
	task_status	VARCHAR(30)
	acceptance_result ?	VARCHAR(20)
FK	accepted_by_employee_id ?	BIGINT > employee
	accepted_at ?	DATETIME

维修工时
repair_labor

PK	**repair_labor_id**	BIGINT
FK	repair_task_id	BIGINT > repair_task
	repair_date	DATE
	duration_hours	DECIMAL(8,2)
	handling_result	VARCHAR(500)
	note ?	VARCHAR(500)
FK	repair_rate_id ?	BIGINT > repair_rate
	hourly_rate_snapshot ?	DECIMAL(12,2)
	labor_amount ?	DECIMAL(12,2)
	submitted_at ?	DATETIME
FK	contractor_id	BIGINT > repair_contractor

维修承包商
repair_contractor

PK	**contractor_id**	BIGINT
UQ	contractor_number	VARCHAR(30)
	contractor_name	VARCHAR(120)
	contact_name ?	VARCHAR(100)
UQ	email	VARCHAR(255)
	phone ?	VARCHAR(40)
	business_status	VARCHAR(20)
	address ?	VARCHAR(255)

调拨与排班

员工班次
employee_shift

PK	**shift_id**	BIGINT
UQ	shift_number	VARCHAR(30)
	shift_date	DATE
FK	store_id	BIGINT > store
FK	employee_id	BIGINT > employee
	start_time	TIME
	end_time	TIME
	shift_status	VARCHAR(20)
FK	changed_by_employee_id	BIGINT > employee
	changed_at	DATETIME
	cancellation_reason ?	VARCHAR(255)

调拨
equipment_transfer

PK	**transfer_id**	BIGINT
UQ	transfer_number	VARCHAR(30)
	transfer_type	VARCHAR(20)
FK	source_store_id	BIGINT > store
FK	destination_store_id	BIGINT > store
FK	initiated_by_employee_id	BIGINT > employee
FK	approved_by_employee_id ?	BIGINT > employee
	initiated_at	DATETIME
	approved_at ?	DATETIME
	completed_at ?	DATETIME
	transfer_status	VARCHAR(20)

调拨项目
equipment_transfer_item

PK	**transfer_item_id**	BIGINT
FK	transfer_id	BIGINT > equipment_transfer
FK	equipment_id	BIGINT > equipment
	dispatch_result	VARCHAR(20)
FK	dispatched_by_employee_id ?	BIGINT > employee
	dispatched_at ?	DATETIME
	receipt_result	VARCHAR(20)
FK	received_by_employee_id ?	BIGINT > employee
	received_at ?	DATETIME
	exception_note ?	VARCHAR(500)

UQ (transfer_id, equipment_id)

维修费率与服务费用

维修服务费用
repair_service_charge

PK	**repair_service_charge_id**	BIGINT
FK UQ	repair_task_id	BIGINT > repair_task
	total_hours	DECIMAL(8,2)
	service_amount	DECIMAL(12,2)
	charge_status	VARCHAR(20)
FK	confirmed_by_employee_id ?	BIGINT > employee
	confirmed_at ?	DATETIME

维修费率
repair_rate

PK	**repair_rate_id**	BIGINT
FK	contractor_id	BIGINT > repair_contractor
	hourly_rate	DECIMAL(12,2)
	effective_from	DATE
	effective_to ?	DATE
	created_at	DATETIME